

Setting Up Colab with Your Google Account

Colab is a free cloud-based notebook environment that requires no setup to use that can be accessed from any browser/device, while providing free access to computing resources including GPUs and TPUs.

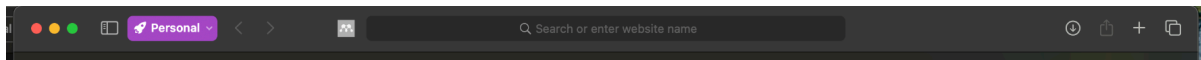
This document outlines the straightforward steps to begin using Google Colaboratory (Colab) with your existing Google account.

Accessing Colab

To access Colab, you simply need a web browser and an internet connection.

Step 1: Open Your Web Browser

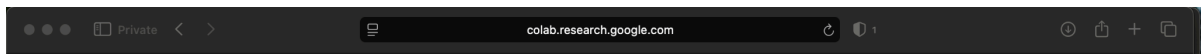
Launch your preferred web browser (e.g., Chrome, Firefox, Safari)

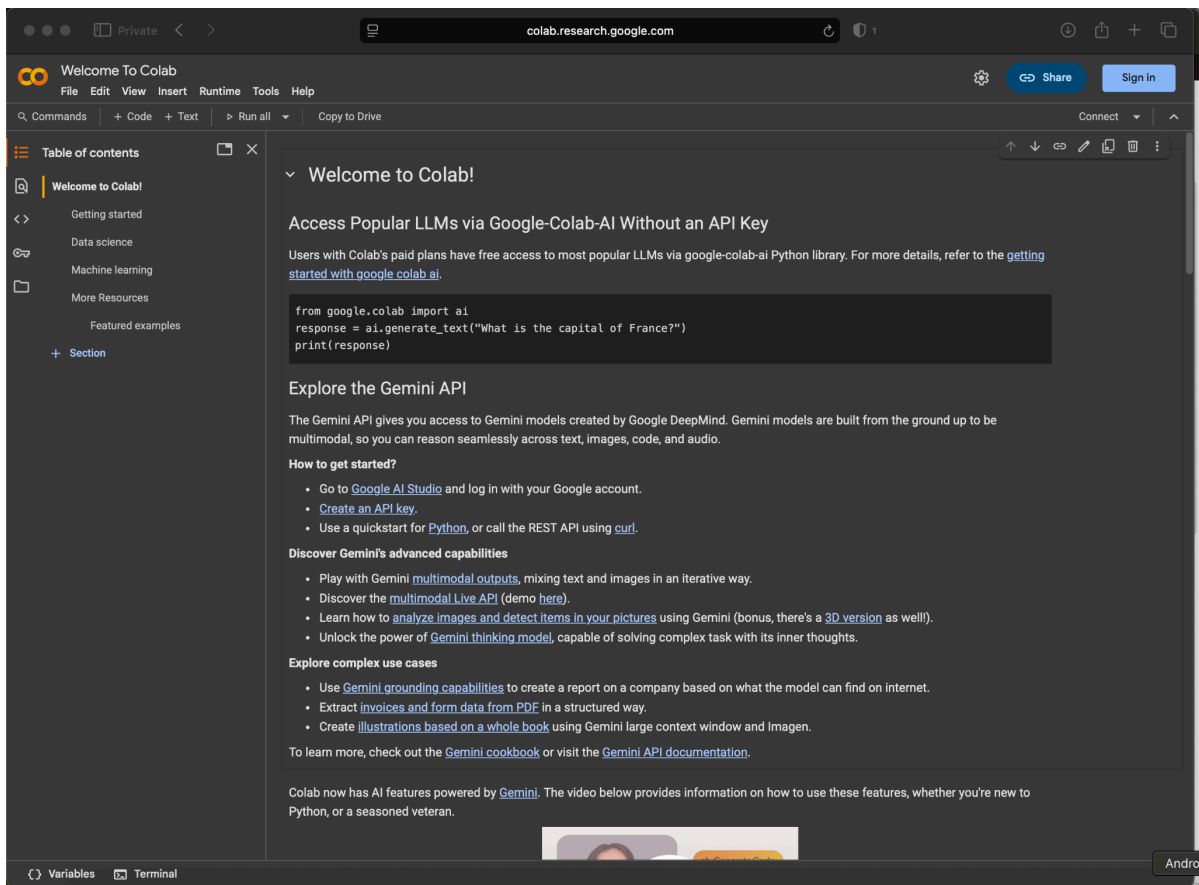


Step 2: Navigate to Colab

In the address bar, type `colab.research.google.com` and press Enter.

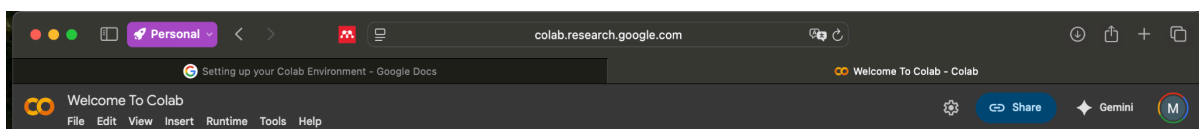
This will take you to the Colab homepage. (If you're already signed in a dialog asking you to choose/create a notebook would open up)



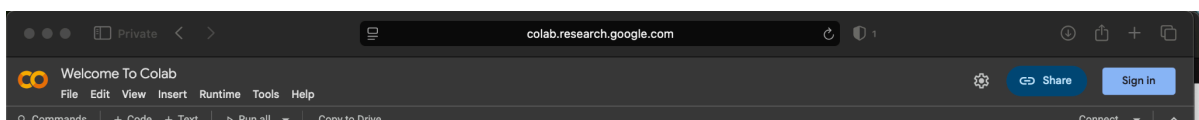


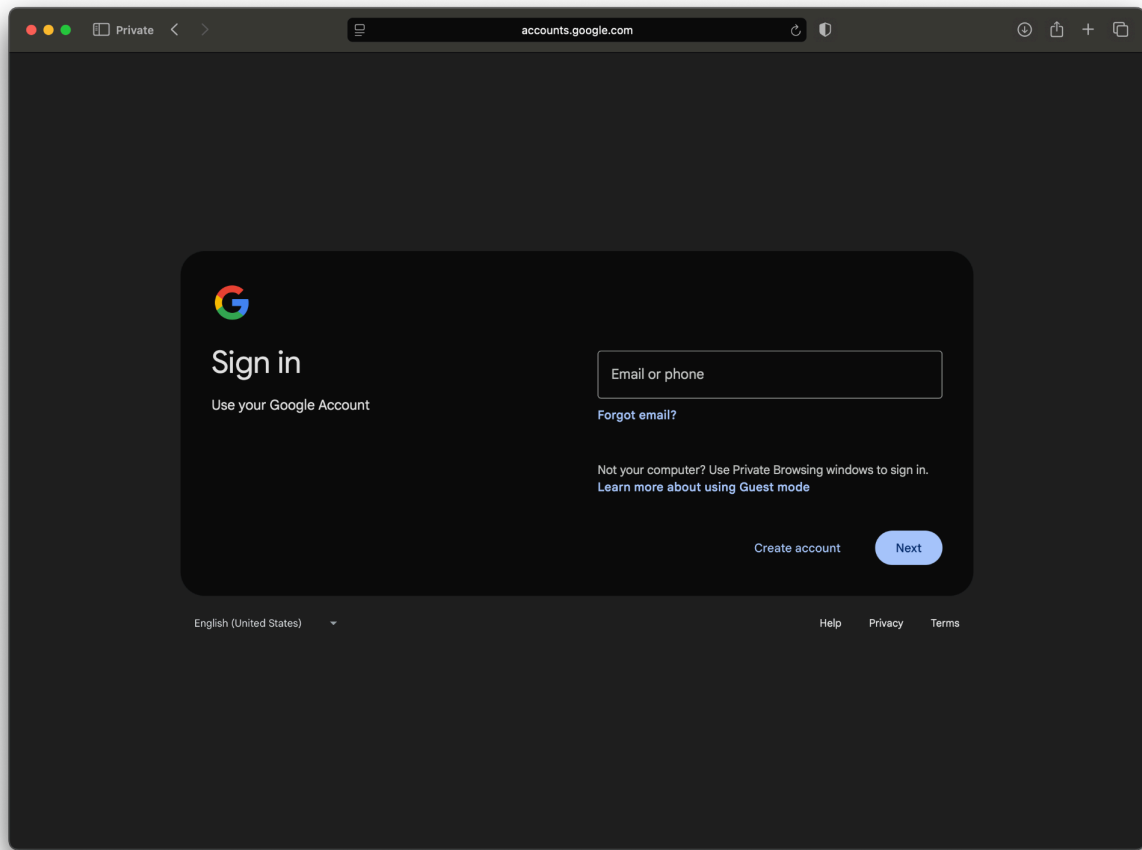
Step 3: Sign In with Your Google Account (Optional; If you're not logged in)

If you're signed in, you would see your profile photo or username's first initial on the top right corner of the window.

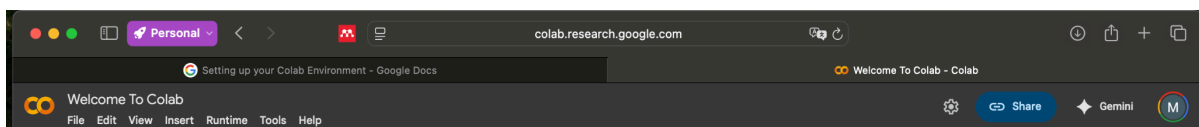


If you're not signed in, you should see an option on the top right corner of your screen. Click on that to complete the process and sign in





Put in your email and password as you normally would sign in and you should successfully get signed in.

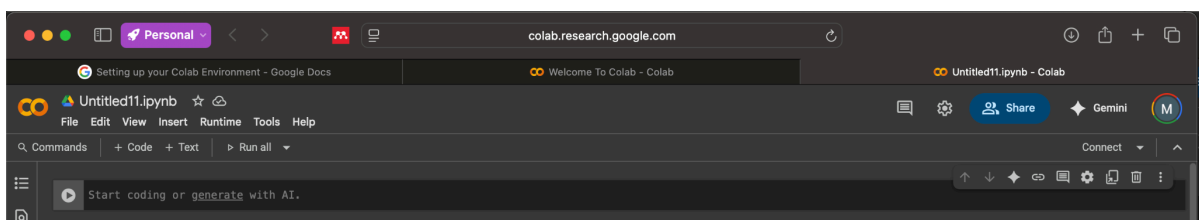
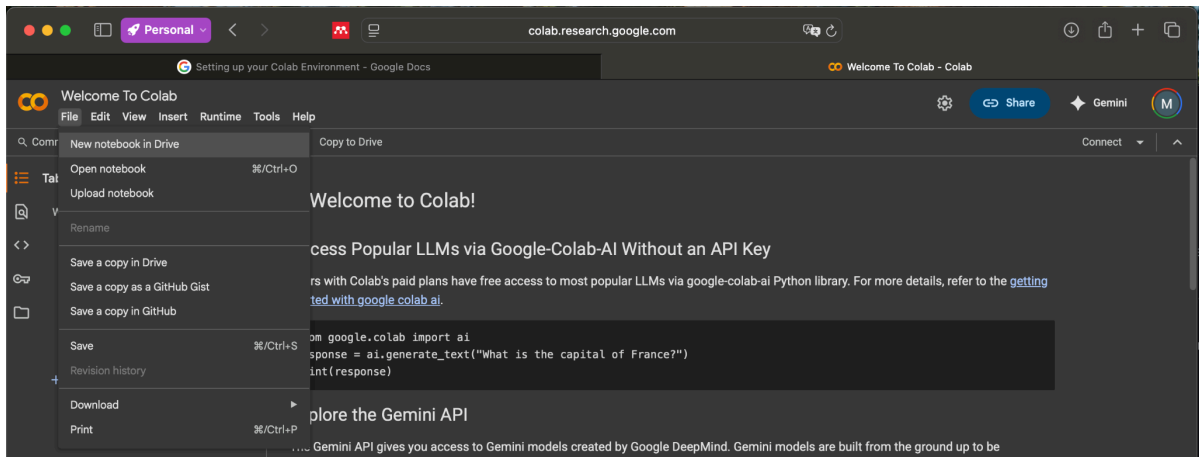


Creating Your First Colab Notebook

Once you've accessed the Colab homepage, you can immediately start creating and working with notebooks.

Option 1: New Notebook

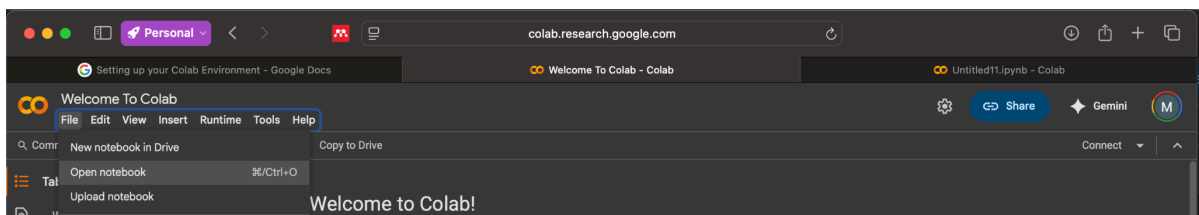
On the Colab homepage, under the File menu button in the top right corner you should see "New notebook in Drive". Click on this to open a blank Colab notebook.



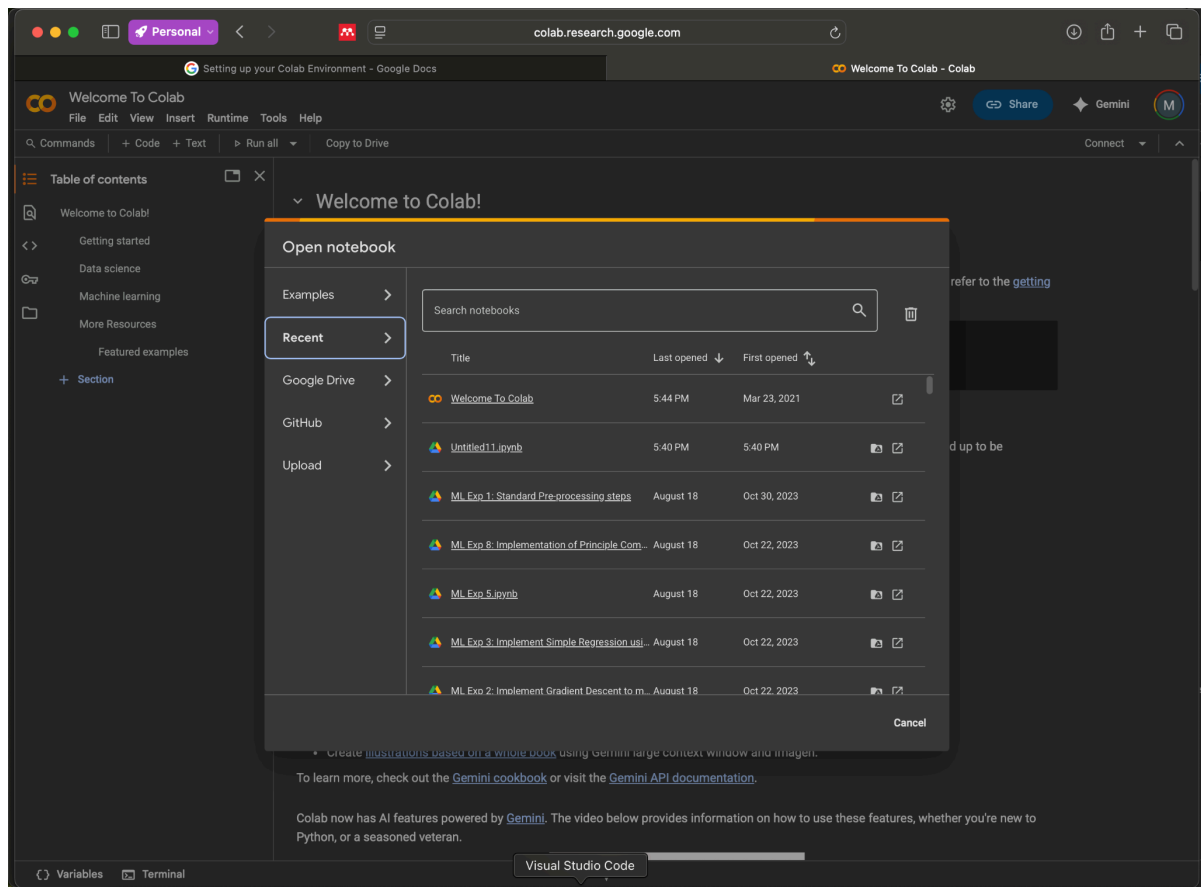
Option 2: Open Notebook from Google Drive

If you have existing notebooks on your Google Drive, you can open them directly.

From the Colab homepage, navigate to the "File" menu and select "Open notebook". This will allow you to browse your Google Drive for `.ipynb` files.



A dialog box shown as follows shall appear on Colab



As an example let's open the notebook "ML Exp 1: Standard Pre-processing steps"

The screenshot shows a Google Colab notebook titled "ML Exp 1: Standard Pre-processing steps". The code in the notebook performs the following steps:

- Load and import required libraries: `import pandas as pd`, `from sklearn.preprocessing import StandardScaler, MinMaxScaler`, and `from sklearn.impute import SimpleImputer`.
- Load the data to be pre-processed into a dictionary `data` with the following structure:

```
data = {
    "CustomerID": [1, 2, 3, 4, 5],
    "Age": [30, 22, 40, None, 25],
    "Income": [50000, None, 60000, 75000, 80000],
    "PurchaseHistory": [10, 5, 8, 12, None],
    "Categorical_column": ["Low", "Low", "Medium", "High", "High"]
}
```
- Create a `pandas.DataFrame` object `df` from the dictionary.
- Display the dataframe, showing a table with 5 rows and 5 columns: `CustomerID`, `Age`, `Income`, `PurchaseHistory`, and `Categorical_column`.
- Perform imputation using `SimpleImputer` with the strategy "mean" for the `Age` and `Income` columns.
- Display the dataframe again, showing the imputed values for `Age` and `Income`.

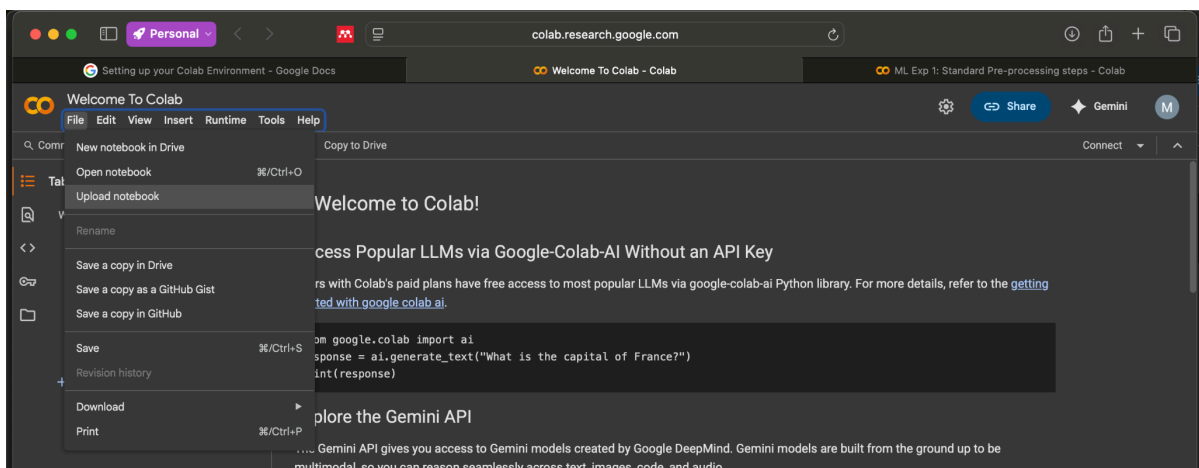
CustomerID	Age	Income	PurchaseHistory	Categorical_column	
0	1	30.0	50000.0	10.0	Low
1	2	22.0	NaN	5.0	Low
2	3	40.0	60000.0	8.0	Medium
3	4	NaN	75000.0	12.0	High
4	5	25.0	80000.0	NaN	High

CustomerID	Age	Income	PurchaseHistory	Categorical_column	
0	1	30.00	50000.0	10.0	Low
1	2	22.00	66250.0	5.0	Low
2	3	40.00	60000.0	8.0	Medium
3	4	29.25	75000.0	12.0	High
4	5	25.00	80000.0	NaN	High

Option 3: Upload Notebook

You can also upload a notebook from your local machine.

Go to "File" > "Upload notebook"



Select the `.ipynb` file you wish to upload.

Saving Your Work

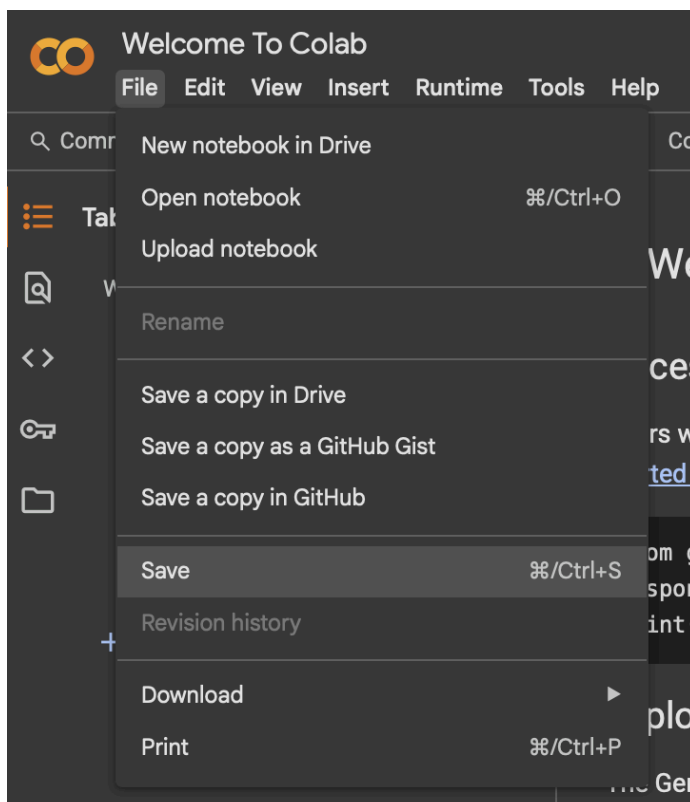
Colab notebooks are automatically saved to your Google Drive.

- **Automatic Saving:**

Colab automatically saves your changes as you work. You can see the "Saved" status at the top of the notebook interface.

- **Manual Saving:**

You can also manually save your notebook by going to "File" > "Save"



Additional Resources/Quick Access Links

- **Colab Welcome Page:** <https://colab.research.google.com>
- **Colab Documentation:** <https://cloud.google.com/colab/docs>
- **Colab Community Forum:**
Reddit Forum: <https://www.reddit.com/r/GoogleColab>
DEV Community: <https://dev.to/t/googlecolab>

Next Steps

Now that you have your Colab environment set up, you can start exploring its features, running Python code, and leveraging its powerful capabilities for data science, machine learning, and more.

If you encounter any issues during setup, please refer to the Colab help documentation or seek assistance from online communities.